

Simon Ellershaw

 Email  Webpage  LinkedIn  Twitter  Github

Education

PhD Candidate, University College London, AI-Enabled Healthcare, MRes Avg Mark 79% 2020-Present

- **Fully funded** UKRI studentship at UCL's Centre of Doctoral Training in AI-Enabled Healthcare Systems
- Awarded PhD placement at **Alan Turing Institute** (UK's National Institute for Data Science and AI)
- PhD thesis focuses on
 1. **Inference:** LLM generated discharge summaries, guided by clinical guidelines and evaluated by clinicians
 2. **Pre-training: UK's first national AI healthcare model** trained on **57 million electronic health records**
 3. **Post-training:** Using real-world patient outcomes as a verifiable reward for reinforcement learning
- Evaluated **vision transformers** for explainable medical imaging diagnosis in collaboration with Moorfield's Eye Hospital
- Collaborations across NHS hospitals and placements that have deepened my **healthcare domain knowledge**
- Completed advanced modules, including **deep learning, computer vision** and **natural language processing**

MSc, Imperial College London, Computing Science, Distinction, Avg Mark 83% 2019-2020

- Gained a distinction for thesis on '3D Infant **Pose Estimation** using Transfer Learning' from **video** input
- Awarded **best group project** for developing a gamified football penalty app using **augmented reality** and **computer vision**. The R&D from the project went on to be integrated into a [commercially released app](#) 
- Achieved first-class attainment in all modules, including **algorithms, computer architecture** and **databases**

BSc, Durham University, Natural Sciences, 1st Class, Avg Mark 76% 2016-2019

- Studied advanced concepts in Computer Science, Physics and Biochemistry
- Delivered high-quality scientific **reports, presentations** and academic **literature reviews**

The King's School Chester A Level: 3 A*'s (Physics, Biology, Chemistry), 1 A (Maths) GCSE: 8 A*'s, 2 A 2009-2016

Industry Experience

Parexel AI Labs, NLP Engineer II, San Francisco Sep 2021-Present

12-month internship at the AI lab of one of the world's largest clinical trials companies. Continue now as a Level 2 NLP engineer 1 day a week during PhD studies.

- One of 8 participants on the prestigious **Silicon Valley Internship Programme**
- Trained, evaluated and deployed full-stack **AI solutions** into production for pharmacovigilance and quality assurance
- Integrated **transformers** and **LLM** implementations into in-house NLP stack
- Lead collaborator with **Stanford** and **University of Ghent** to develop a public pharmacovigilance dataset

Metaswitch, Software Developer Intern, London Jul 2019-Sep 2019

8-week technology internship at a market-leading telecoms company.

- Designed solutions based on **cloud-native technologies**, including Docker
- Recognized for **strong communication skills**, an **admirable inquisitive streak**, and **openness to feedback**, resulting in a graduate job offer

Motor Design Ltd, Software Developer Intern, Wrexham Jul 2018-Sep 2018

12-week software development internship at an electric motor design software startup.

- Led the development of an automated reporting system and presented results to the President and CEO

Technical Skills

- Languages: **Python (advanced)** including libraries such as **PyTorch, NumPy, Pandas**
- Experience in **multi-GPU training** and **AI evaluation**

Hobbies

- Swimming, Rowing and Water Polo- Represented Durham and Imperial University at a national level
- Hiking, Surfing, and Cycling- I have a passion for exploring the outdoors, self-organising trips to Asia, Europe and the USA

Research Projects

Foresight- The UK's First National AI Healthcare Model

A generative auto-regressive transformer model trained from scratch on 57 million de-identified coded NHS England patient records.

- **Press Coverage:** [UCL](#) [Nature](#) [BBC Radio](#) [New Scientist](#) [The Independent](#) [The Telegraph](#) [The Mirror](#) [Fox News](#)
- **Presentations:** [Healtac 2025](#) [UK AI 2025](#)
- **Posters:** [UK Generative AI Hub Launch](#) [3 Minute Thesis \(Departmental Prize\)](#)

Automated Discharge Summaries

Developing a system to automate discharge summaries using large language models guided by clinical guidelines. Completed a proof of concept using publicly available data, and currently conducting a retrospective study within a UCLH.

- **Publication:** “Automated Generation of Hospital Discharge Summaries using Clinical Guidelines and Large Language Models”, AAAI 2024 Spring Symposium on Clinical Foundation Models, 2023 [Paper](#) [Poster](#)
- **Presentations:** [Healtac 2024](#) [Manchester University UCL MSC Health Informatics Guest Lecture](#)
- **Resources:** [Github Repository](#)

Diabetic Retinopathy Vision Transformer

Applying vision transformers to the task of diabetic retinopathy classification (and comparing to CNNs). In collaboration with Moorfield's Eye Hospital.

- **Posters:** [UCL AI Enabled Healthcare Annual Symposium, 2021. Awarded Best Poster Prize](#)
- **Resources:** [Github Repository](#)

3D Infant Pose Estimation

Development of a deep learning model for 3D infant pose estimation, outperforming existing methods without relying on depth data. This work lays the foundation for automated movement assessment tools that could aid in early diagnosis of neurological disorders like cerebral palsy.

- **Publication:** “3D Infant Pose Estimation Using Transfer Learning”, Medical Imaging meets NeurIPS workshop, 2020. Acceptance rate for orals 14.5% [Abstract](#) [Oral Presentation](#)
- **Resources:** [Github Repository](#)

Collaborations

- “BioDEX: Large-Scale Biomedical Adverse Drug Event Extraction for Real-World Pharmacovigilance”, Accepted EMNLP Proceedings, 2023 [Paper](#)
- “Unsupervised Human Pose Estimation through Transforming Shape Templates”, CVPR, 2021 [Paper](#) [Project Site](#)
- “A new meaning for NLP – the trials and tribulations of natural language processing with GPT-3 in ophthalmology”, British Journal of Ophthalmology, 2021 [Paper](#)

Invited Talks

- [Generative AI in the NHS](#), MSD Pharmaceuticals, London, 2025
- [Introduction to Large Language Models](#), Artificial Intelligence in Healthcare, UCL Medical School, 2025
- [AI in Healthcare: Is it all hype?](#) [Ulfasa Meeting on Education and Leadership in Palliative Care](#), Emmanuel College, Cambridge, 2024